

Problem Set 1

Spring 2026

Due: 11:59 pm Monday, 9 Feb.

To facilitate grading and timely feedback
please note that all submissions are through Gradescope.

Solve each problem on a new page.

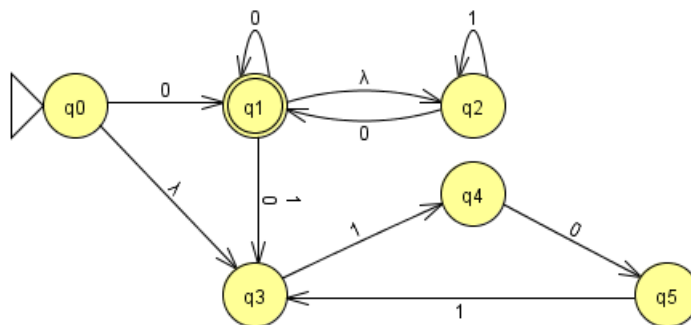
Please clearly identify collaborator names and the resources used for each problem.

- Using his intuition that languages accepted by finite automata require remembering only a finite pattern, Sheikh Chilly would like to argue that the following language L_n is not regular. Obviously, he argues, remembering multiples of a given integer n requires unbounded memory. For the alphabet $\Sigma = \{0, 1\}$, let the language L_n be given by

$$L_n = \{w \in \Sigma^* \mid w \text{ is a binary number that is a multiple of } n\},$$

for $n \geq 1$. Show that for all valid choices of n , the language L_n is regular.

- Convert the following NFA into a DFA.



- Let $\Sigma = \{a, b\}$ and $L_k = \{ww : w \in \Sigma^* \text{ and } w \text{ is of length } k\}$. Show that for each k , no DFA can recognize L_k with fewer than 2^k states.
- For languages A and B , let the *interleave* of A and B be the language

$$\{w \mid w = a_1b_1 \dots a_kb_k, \text{ where } a_1 \dots a_k \in A \text{ and } b_1 \dots b_k \in B, \text{ each } a_i, b_i \in \Sigma\}.$$

Show that the class of regular languages is closed under *interleave*.

- Let $L = \{1^n : n \text{ is composite}\}$. Show that L is not regular.